B. Tech. (3rd Sem) (Common for All Streams except CSE-Industry Integrated)
BCSE-501 (Data Structure & Algorithms)

L	T	P		
3	0	0	**Continuous evaluation**	**40**
			End semester exam	**60**
			Total marks	**100**
			Credits	**3.0**

Course Objectives:

1. Develop and analyse algorithms for linked list.
2. To introduce the concepts of Tree and Graph and their traversal algorithms
3. Implement various Hashing and Collision Resolution Technique
4. To introduce various techniques for representation of the data in the real world

UNIT-I

Re-explore Linear Data Structure, Static and Dynamic Memory Management

Linked Lists: Introduction, singly linked list, representation of a linked list in memory, operations on a single linked list; Applications of linked lists: Polynomial representation and sparse matrix manipulation. Types of linked lists: Circular linked lists, doubly linked lists; Linked list representation and operations of Stack, linked list representation and operations of queue

UNIT-II

Trees: Basic concept, binary tree, binary tree representation, array and linked representations, binary tree traversal, binary tree variants, threaded binary trees, application of trees.

UNIT-III

Binary search trees: Binary search trees, properties, and operations; Balanced search trees: AVL trees; Introduction to M-Way search trees, B trees;

UNIT-IV

Graphs: Basic concept, graph terminology, Graph Representations - Adjacency matrix, Adjacency lists, graph implementation, Graph traversals – BFS, DFS.

Hashing and collision: Introduction, hash tables, hash functions, collisions, applications of hashing.

Course Outcomes: After completion of this course, student will be able to:

i) Select appropriate data structures as applied to specified problem definition.
ii) Implement Linked list data structures.
iii) Implement appropriate technique of tree and graph for given problem.
iv) Implement real life applications using Hashing and Collision Resolution Technique.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. Ellis Horowitz and Sartaj Sahni, “Fundamentals of Data Structures”, Illustrated Edition, Computer Science Press.
2. Seymour Lipschutz, “Data Structures”, Schaum's Outline Series, McGraw Hill Publication.

Reference Books:

1. R. L. Kruse, B.P. Leary and C.L Tondo, “Data structure and program design in C”, PHI.
2. A. V. Aho, J. E. Hopcroft and T. D. Ullman, “Data Structures and Algorithms”, Addison-Wesley.

B. Tech. (3rd Sem) (Common for All Streams except CSE-Industry Integrated)
BCSE-501L (Data Structure & Algorithms Lab)

L	T	P		
0	0	2	**Continuous evaluation**	**40**
			End semester exam	**60**
			Total marks	**100**
			Credits	**1.0**

Course Objectives:

1. To assess how the choice of data structures and algorithm design methods impacts the performance of programs.
2. To choose the appropriate data structure and algorithm design method for a specified application.
3. To solve problems using data structures such as linear lists, stacks, queues, binary trees, heaps binary search trees, and graphs and writing programs for these solutions.

List of Practical

1. Declare a 2D Array with dimensions of 9 x 9. Implement the following operations on this array:
   i) Search ii) Traversal iii) Sum of all elements iv) Insertion v) Deletion.

Use the learned concept of this practical to solve **Su-Do-Ku Puzzle** whose statement is:
Su-Do-Ku Puzzle –
A Sudoku is a problem where there are is an incomplete 9 x 9 table of numbers which must be filled according to several rules:

ü Within any of the 9 individual 3 x 3 boxes, each of the numbers 1 to 9 must be found.
ü Within any column of the 9 x 9 grid, each of the numbers 1 to 9 must be found.
ü Within any row of the 9 x 9 grid, each of the numbers 1 to 9 must be found.

2. Implement representation of sparse matrix using array and link list.

Use the learned concept of this practical to solve **Conway Game of Life** whose statement is:
Conway Game of Life – The project introduces the learner to the challenge of implementing a two-dimensional array when the vast majority of the elements are empty, or they are zero. This has the learners explore how best to implement a sparse matrix, tests the implementation using the Conway's Game of Life and determines which is best.

3. Implement various operations of Stack.

Use the learned concept of this practical to solve **Tower of Hanoi** problem (with & without recursion) whose statement is:
Tower of Hanoi – The Tower of Hanoi is a mathematical problem which consists of three rods and multiple disks. Initially, all the disks are placed on one rod, one over the other in ascending order of size similar to a cone-shaped tower. The objective of this problem is to move the stack of disks from the initial rod to another rod, following these rules:

- A disk cannot be placed on top of a smaller disk.
- No disk can be placed on top of the smaller disk.

4. Implement various operations of Queue.

Use the learned concept of this practical to implement job scheduling algorithms used to solve **Job scheduling problem** whose statement is:
Job Scheduling – Job scheduling is the process of allocating system resources to many different tasks / jobs by an operating system (OS). The system handles prioritized job queues that are waiting CPU time and it should determine which job to be taken from which queue and the amount of time to be allocated for the job.

5. Implement memory representation of binary tree using array and link list.

6. Represent a mathematical expression in the form of binary tree & evaluate that mathematical expression.
   Use the learned concept of 5th & 6th practical to make

Expression Calculator which is used for Converting infix to postfix expressions and evaluating results using trees data structure.

7. Implement memory representation of given location in graph.

Use the learned concept of this practical to solve **Travelling salesman problem** which is the challenge of finding the shortest yet most efficient route for a person to take given a list of specific destinations.

Integrated Project (Mandatory) based upon the learnt concepts

Dictionary Management – In this project, students will be required to read a dictionary file into different data structures (Array, Linked List, Stack, Queue) to perform various operations (Search, Insert (with sorting), Delete, View) and improve searching using hashing techniques.

Course Outcomes: After completion of this course, students will be able to:

1. Write well-structured procedure-oriented programs.
2. Implement the Stack ADT using both array based and linked-list based data structures.
3. Implement the Queue ADT using both array based circular queue and linked-list based implementations.
4. Implement binary search trees and graphs.